# Fourier Series

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***Abstract*—This manual provides a simple introduction to Fourier Series and its application through filter in a basic charger circuit.**

## 1 Periodic Function

Let

$$x(t) = A_0 \left|\sin\left(2\pi f_0 t\right)\right| \tag{1.1}$$

1.1 Plot $x(t)$.

*Solution.* The plot of $x(t)$ is shown in Fig. 1.1.

1.2 Show that $x(t)$ is periodic and find its period.

*Solution.* Note that,

$$x\left(t+\frac{1}{2f_0}\right) = A_0\left|\sin\left(2\pi f_0\left(t+\frac{1}{2f_0}\right)\right)\right| \tag{1.2}$$
$$= A_0\left|\sin\left(2\pi f_0 t+\pi\right)\right| \tag{1.3}$$
$$= A_0\left|-\sin\left(2\pi f_0 t\right)\right| \tag{1.4}$$
$$= A_0\left|\sin\left(2\pi f_0 t\right)\right| \tag{1.5}$$
$$= x(t). \tag{1.6}$$

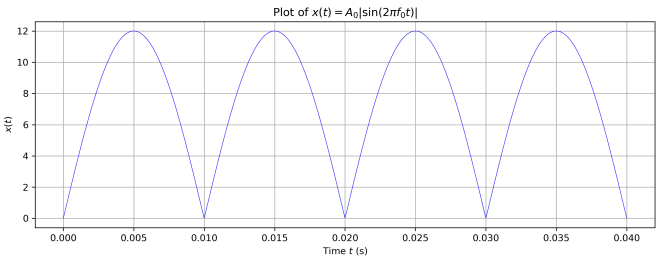



Fig. 1.1. Plot of $x(t) = A_0\left|\sin(2\pi f_0 t)\right|$.

Hence, $x(t)$ is periodic with fundamental period

$$\boxed{T_0 = \frac{1}{2f_0}}. \tag{1.7}$$

## 2 Fourier Series

Consider $A_0 = 12$ and $f_0 = 50$ for all numerical calculations.

2.1 If

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \tag{2.1}$$

show that

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-\mathrm{j}2\pi k f_0 t}\, dt. \tag{2.2}$$

*Solution.* For some $n \in \mathbb{Z}$, multiplying both sides of (2.1) by $e^{-\mathrm{j}2\pi n f_0 t}$ gives

$$x(t)e^{-\mathrm{j}2\pi n f_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi(k-n) f_0 t}. \tag{2.3}$$

Now, integrating both sides over the interval $\left[-\frac{1}{2f_0}, \frac{1}{2f_0}\right]$,

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)e^{-\mathrm{j}2\pi nf_0t}\,dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi(k-n)f_0t}\,dt. \tag{2.4}$$

The integral on the right can be evaluated using the orthogonality property of complex exponentials:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi(k-n)f_0t}\,dt = \begin{cases} \dfrac{1}{f_0}, & k = n, \\ 0, & k \neq n. \end{cases} \tag{2.5}$$

Note that the integration interval has length $1/f_0$. Although the fundamental period of the given signal $x(t) = A_0|\sin(2\pi f_0 t)|$ is $1/(2f_0)$, the chosen Fourier basis $e^{\mathrm{j}2\pi kf_0t}$ has period $1/f_0$. Thus, the integration interval covers two fundamental periods of the signal and one period of the Fourier basis.
Consequently, all terms in the summation vanish except the term for which $k = n$. Therefore,

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)e^{-\mathrm{j}2\pi nf_0t}\,dt = \frac{c_n}{f_0}. \tag{2.6}$$

Hence,

$$\boxed{c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)e^{-\mathrm{j}2\pi nf_0t}\,dt}. \tag{2.7}$$

2.2 Find $c_k$ for (1.1).
*Solution.* Using (2.2),

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} A_0\,|\sin(2\pi f_0 t)|\,e^{-\mathrm{j}2\pi nf_0t}\,dt. \tag{2.8}$$

Using Euler's identity,

$$e^{-\mathrm{j}\theta} = \cos\theta - \mathrm{j}\sin\theta, \tag{2.9}$$

we obtain

$$c_n = f_0A_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} |\sin(2\pi f_0t)|\cos(2\pi nf_0t)\,dt - \mathrm{j}f_0A_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} |\sin(2\pi f_0t)|\sin(2\pi nf_0t)\,dt. \tag{2.10}$$

Since $x(t)$ is even and the sine term is odd, the imaginary part vanishes. Thus,

$$c_n = 2f_0A_0 \int_0^{\frac{1}{2f_0}} \sin(2\pi f_0t)\cos(2\pi nf_0t)\,dt. \tag{2.11}$$

Using the identity

$$2\sin A\cos B = \sin(A+B) + \sin(A-B), \tag{2.12}$$

we get

$$c_n = f_0A_0 \int_0^{\frac{1}{2f_0}} \sin(2\pi(n+1)f_0t)\,dt + f_0A_0 \int_0^{\frac{1}{2f_0}} \sin(2\pi(1-n)f_0t)\,dt. \tag{2.13}$$

For $n \neq \pm 1$, evaluating the integrals gives

$$c_n = \frac{A_0}{2\pi}\left[\frac{1-(-1)^{n+1}}{n+1} + \frac{1-(-1)^{1-n}}{1-n}\right] = \frac{A_0(1+(-1)^n)}{\pi(1-n^2)}. \tag{2.14}$$

For $n = \pm 1$, direct evaluation of the original integral gives $c_n = 0$.
Therefore,

$$\boxed{c_n = \begin{cases} \dfrac{2A_0}{\pi(1-n^2)}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}} \tag{2.15}$$

Here, $n = 0$ is included in the even case, giving $c_0 = 2A_0/\pi$.

2.3 Verify (2.1) using Python.
*Solution.* The Python code `codes/2_3.py` verifies (2.1) by comparing the original signal with its Fourier-series reconstruction, as shown

in Fig. 2.3.

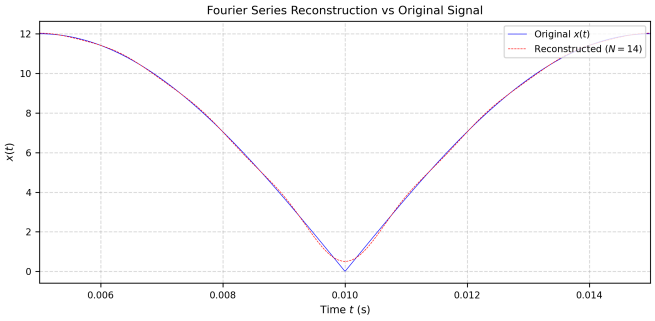



Fig. 2.3. Verification of the complex Fourier series.

2.4 Show that

$$x(t) = a_0 + \sum_{k=1}^{\infty} \left[a_k \cos(2\pi k f_0 t) + b_k \sin(2\pi k f_0 t)\right] \tag{2.16}$$

and obtain the formulae for $a_k$ and $b_k$.
*Solution.* From (2.1),

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \tag{2.17}$$

$$= c_0 + \sum_{k=1}^{\infty} \left[c_k e^{\mathrm{j}2\pi k f_0 t} + c_{-k} e^{-\mathrm{j}2\pi k f_0 t}\right]. \tag{2.18}$$

Using Euler's identities,

$$e^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta, \tag{2.19}$$

$$e^{-\mathrm{j}\theta} = \cos\theta - \mathrm{j}\sin\theta, \tag{2.20}$$

we obtain

$$x(t) = c_0 + \sum_{k=1}^{\infty} \Big[(c_k + c_{-k})\cos(2\pi k f_0 t) + \mathrm{j}(c_k - c_{-k})\sin(2\pi k f_0 t)\Big]. \tag{2.21}$$

Comparing this with (2.16), we get

$$a_k = \begin{cases} c_0, & k = 0, \\ c_k + c_{-k}, & k > 0, \end{cases} \tag{2.22}$$

and

$$b_k = \mathrm{j}(c_k - c_{-k}), \qquad k > 0. \tag{2.23}$$

Using the coefficient formula (2.2), the corresponding integral expressions are

$$\begin{aligned} a_0 &= f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\,dt, \\ a_k &= 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\cos(2\pi k f_0 t)\,dt, \quad k > 0, \\ b_k &= 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\sin(2\pi k f_0 t)\,dt, \quad k > 0. \end{aligned} \tag{2.24}$$

2.5 Find $a_k$ and $b_k$ for (1.1).
*Solution.* Since $x(t)$ is an even function,

$$x(-t) = x(t). \tag{2.25}$$

From the complex Fourier-series representation,

$$x(-t) = \sum_{k=-\infty}^{\infty} c_k e^{-\mathrm{j}2\pi k f_0 t} \tag{2.26}$$

$$= \sum_{k=-\infty}^{\infty} c_{-k} e^{\mathrm{j}2\pi k f_0 t}. \tag{2.27}$$

Comparing this with

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t}, \tag{2.28}$$

we obtain

$$c_k = c_{-k}. \tag{2.29}$$

Therefore, from (2.22) and (2.23),

$$a_k = \begin{cases} \dfrac{2A_0}{\pi}, & k = 0, \\ \dfrac{4A_0}{\pi(1-k^2)}, & k > 0,\ k \text{ even}, \\ 0, & k > 0,\ k \text{ odd}, \end{cases} \tag{2.30}$$

$$b_k = 0, \qquad k > 0. \tag{2.31}$$

For $A_0 = 12$, these become

$$a_0 = \frac{24}{\pi}, \tag{2.32}$$

$$a_k = \begin{cases} \dfrac{48}{\pi(1-k^2)}, & k > 0,\ k \text{ even}, \\ 0, & k > 0,\ k \text{ odd}, \end{cases} \tag{2.33}$$

$$b_k = 0. \tag{2.34}$$

2.6 Verify (2.16) using Python.
*Solution.* The Python code `codes/2_6.py` verifies (2.16) by comparing the original signal with its real Fourier-series reconstruction, as shown in Fig. 2.6.

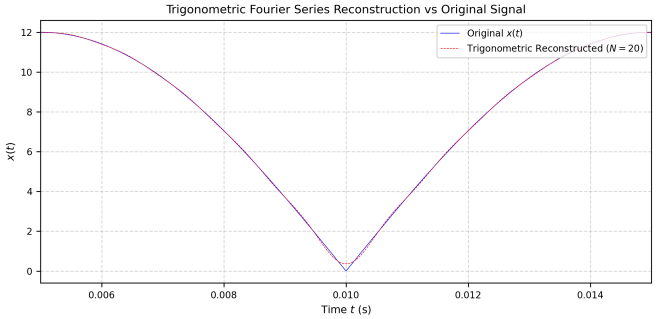



Fig. 2.6. Verification of the real Fourier series.

## 3 Fourier Transform

We use the Fourier-transform convention

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-\mathrm{j}2\pi f t}\, dt, \tag{3.1}$$

with inverse transform

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{\mathrm{j}2\pi f t}\, df. \tag{3.2}$$

3.1 The unit impulse satisfies

$$\delta(t) = 0, \quad t \neq 0, \tag{3.3}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1. \tag{3.4}$$

*Solution.* The Dirac delta is zero everywhere except at the origin, but its total integral is unity. More generally, it has the sifting property

$$\int_{-\infty}^{\infty} g(t)\delta(t - t_0)\, dt = g(t_0). \tag{3.5}$$

This property allows an impulse to select the value of a function at its location.

3.2 State the Fourier-transform pair for $g(t)$.
*Solution.* By definition, the Fourier transform of $g(t)$ is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-\mathrm{j}2\pi f t}\, dt. \tag{3.6}$$

The corresponding inverse transform is

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{\mathrm{j}2\pi f t}\, df. \tag{3.7}$$

Hence, we write $g(t) \overset{\mathcal{F}}{\longleftrightarrow} G(f)$.

3.3 Show that

$$g(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} G(f) e^{-\mathrm{j}2\pi f t_0}. \tag{3.8}$$

*Solution.* Let $y(t) = g(t - t_0)$. Its Fourier transform is

$$Y(f) = \int_{-\infty}^{\infty} g(t - t_0) e^{-\mathrm{j}2\pi f t}\, dt. \tag{3.9}$$

Using the substitution $u = t - t_0$, so that $t = u + t_0$ and $dt = du$,

$$Y(f) = \int_{-\infty}^{\infty} g(u) e^{-\mathrm{j}2\pi f(u+t_0)}\, du \tag{3.10}$$

$$= e^{-\mathrm{j}2\pi f t_0} \int_{-\infty}^{\infty} g(u) e^{-\mathrm{j}2\pi f u}\, du \tag{3.11}$$

$$= G(f) e^{-\mathrm{j}2\pi f t_0}. \tag{3.12}$$

Therefore,

$$\boxed{g(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} G(f) e^{-\mathrm{j}2\pi f t_0}}. \tag{3.13}$$

3.4 Show that

$$G(t) \overset{\mathcal{F}}{\longleftrightarrow} g(-f). \tag{3.14}$$

*Solution.* From the inverse Fourier-transform relation,

$$g(-f) = \int_{-\infty}^{\infty} G(\nu) e^{-\mathrm{j}2\pi \nu f}\, d\nu. \tag{3.15}$$

This is precisely the Fourier transform of the function $G(t)$, evaluated at frequency $f$. Thus, by the duality property,

$$\boxed{G(t) \overset{\mathcal{F}}{\longleftrightarrow} g(-f)}. \tag{3.16}$$

3.5 Find the Fourier transform of $\delta(t)$.
*Solution.* Using the definition,

$$\mathcal{F}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t) e^{-\mathrm{j}2\pi f t}\, dt. \tag{3.17}$$

By the sifting property, the integrand is evaluated at $t = 0$:

$$\boxed{\delta(t) \xleftrightarrow{\mathcal{F}} 1}. \tag{3.18}$$

3.6 Find the Fourier transform of $e^{-\mathrm{j}2\pi f_0 t}$.
*Solution.* The standard frequency-shift impulse pair is

$$e^{\mathrm{j}2\pi f_0 t} \xleftrightarrow{\mathcal{F}} \delta(f - f_0). \tag{3.19}$$

Replacing $f_0$ by $-f_0$ gives

$$\boxed{e^{-\mathrm{j}2\pi f_0 t} \xleftrightarrow{\mathcal{F}} \delta(f + f_0)}. \tag{3.20}$$

3.7 Find the Fourier transform of $\cos(2\pi f_0 t)$.
*Solution.* Using Euler's identity,

$$\cos(2\pi f_0 t) = \frac{1}{2}\left(e^{\mathrm{j}2\pi f_0 t} + e^{-\mathrm{j}2\pi f_0 t}\right). \tag{3.21}$$

By linearity and the exponential transform pairs,

$$\boxed{\cos(2\pi f_0 t) \xleftrightarrow{\mathcal{F}} \frac{1}{2}\left[\delta(f - f_0) + \delta(f + f_0)\right]}. \tag{3.22}$$

3.8 Find the Fourier transform of $x(t)$ and plot it. Verify using Python.
*Solution.* The signal is periodic with fundamental frequency $2f_0$. Its complex Fourier-series representation can therefore be written as

$$x(t) = \sum_{m=-\infty}^{\infty} d_m e^{\mathrm{j}2\pi(2mf_0)t}, \tag{3.23}$$

where, from the coefficients obtained in Section 2,

$$d_m = c_{2m} = \frac{2A_0}{\pi(1 - 4m^2)}. \tag{3.24}$$

The Fourier transform of a periodic signal is an impulse train in the frequency domain. Hence,

$$X(f) = \sum_{m=-\infty}^{\infty} d_m \delta(f - 2mf_0) \tag{3.25}$$

$$= \boxed{\sum_{m=-\infty}^{\infty} \frac{2A_0}{\pi(1 - 4m^2)} \delta(f - 2mf_0)}. \tag{3.26}$$

For $A_0 = 12$ and $f_0 = 50$ Hz,

$$X(f) = \sum_{m=-\infty}^{\infty} \frac{24}{\pi(1 - 4m^2)} \delta(f - 100m). \tag{3.27}$$

Python code plots the first few spectral lines and compares the analytical Fourier-series reconstruction with the original signal.

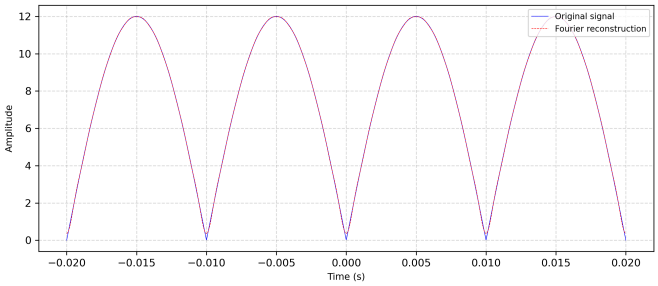



Fig. 3.8. Comparision with the fourier recreation.

3.9 Show that $\operatorname{rect} t \xleftrightarrow{\mathcal{F}} \operatorname{sinc} f$. Verify using Python.
*Solution.* Using the normalized definitions

$$\operatorname{rect} t = \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & |t| > \frac{1}{2}, \end{cases} \tag{3.28}$$

$$\operatorname{sinc} f = \frac{\sin(\pi f)}{\pi f}, \qquad \operatorname{sinc} 0 = 1, \tag{3.29}$$

the Fourier transform is

$$R(f) = \int_{-\infty}^{\infty} \operatorname{rect} t\, e^{-\mathrm{j}2\pi f t}\, dt \tag{3.30}$$

$$= \int_{-1/2}^{1/2} e^{-\mathrm{j}2\pi f t}\, dt \tag{3.31}$$

$$= \left[\frac{e^{-\mathrm{j}2\pi f t}}{-\mathrm{j}2\pi f}\right]_{-1/2}^{1/2} \tag{3.32}$$

$$= \frac{e^{-\mathrm{j}\pi f} - e^{\mathrm{j}\pi f}}{-\mathrm{j}2\pi f} \tag{3.33}$$

$$= \frac{\sin(\pi f)}{\pi f}. \tag{3.34}$$

Therefore,

$$\boxed{\operatorname{rect} t \xleftrightarrow{\mathcal{F}} \operatorname{sinc} f}. \tag{3.35}$$

The following code numerically evaluates the transform and compares it with the analytical sinc function.

3.10 Find the Fourier transform of $\operatorname{sinc} t$. Verify using Python.

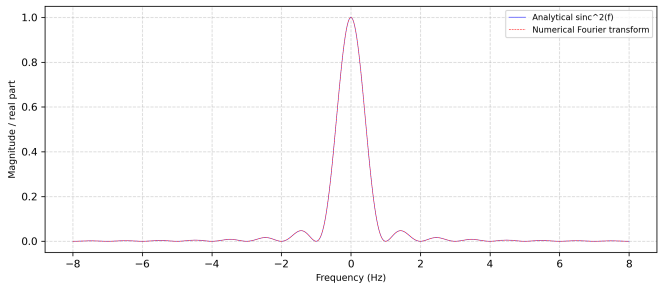



Fig. 3.9. Comparision with the fourier transform.

*Solution.* By the duality property, since

$$\operatorname{rect} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{sinc} f, \tag{3.36}$$

we obtain

$$\operatorname{sinc} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{rect} -f. \tag{3.37}$$

The rectangular function is even, so $\operatorname{rect} -f = \operatorname{rect} f$. Hence,

$$\boxed{\operatorname{sinc} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{rect} f}. \tag{3.38}$$

The following code numerically evaluates the transform over a finite time interval. Increasing the integration range and sampling density improves the approximation. At the discontinuities $f = \pm\frac{1}{2}$, the Fourier-transform value is conventionally taken as the midpoint value $1/2$.

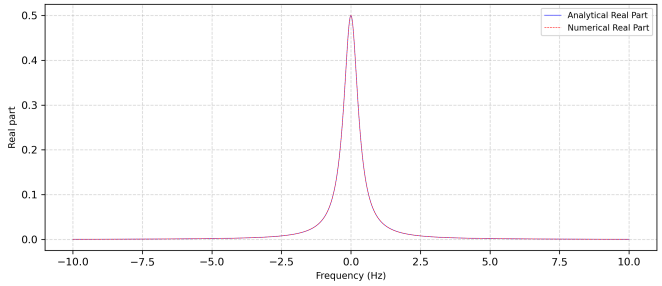



Fig. 3.10. Comparision with the fourier transform.

## 4 Filter

4.1 Find $H(f)$ which transforms $x(t)$ to DC 5V.
4.2 Find $h(t)$.
4.3 Verify your result using through convolution.

## 5 Filter Design

5.1 Design a Butterworth filter for $H(f)$.
5.2 Design a Chebyschev filter for $H(f)$.
5.3 Design a circuit for your Butterworth filter.
5.4 Design a circuit for your Chebyschev filter.

## 6 Circuit

### 6.1 Input Parameters

The practical and reference values used in the following calculations are summarised in Table 4.

### 6.2 Transformer

The primary peak voltage, secondary peak voltage and turns ratio follow from the RMS measurements:

$$V_{p,\mathrm{pk}} = \sqrt{2}V_p = \sqrt{2}(220) \approx 311.1\,\mathrm{V}, \tag{6.1}$$

$$V_{s,\mathrm{pk}} = \sqrt{2}V_s = \sqrt{2}(10.53) \approx 14.89\,\mathrm{V}, \tag{6.2}$$

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{220}{10.53} \approx 20.9. \tag{6.3}$$

The transformer reduces the voltage while retaining the mains frequency and providing galvanic isolation.

### 6.3 Bridge Rectifier

Two diodes conduct during each half-cycle. The peak rectified voltage is therefore

$$V_{o,\mathrm{pk}} = V_{s,\mathrm{pk}} - 2V_D \tag{6.4}$$

$$= 14.89 - 2(0.7) \approx 13.49\,\mathrm{V}. \tag{6.5}$$

For a full-wave rectified sinusoid,

$$V_{o,\mathrm{avg}} = \frac{2V_{o,\mathrm{pk}}}{\pi} \approx 8.59\,\mathrm{V}, \tag{6.6}$$

$$V_{o,\mathrm{rms}} = \frac{V_{o,\mathrm{pk}}}{\sqrt{2}} \approx 9.54\,\mathrm{V}, \tag{6.7}$$

$$f_r = 2f = 100\,\mathrm{Hz}, \qquad T_r = \frac{1}{f_r} = 10\,\mathrm{ms}. \tag{6.8}$$

The approximate PIV is the secondary peak voltage, well below the 1N4007 rating listed in Table 4.

### 6.4 Smoothing Capacitor

The capacitor value used is $C = 2200\,\mu\mathrm{F}$. It charges near the rectified peaks and discharges through the load between peaks. For a resistive load $R_L$, its discharge is governed by

$$C\frac{dv_C(t)}{dt} + \frac{v_C(t)}{R_L} = 0, \tag{6.9}$$

$$v_C(t) = V_{\max} e^{-\frac{t-t_0}{R_L C}}. \tag{6.10}$$

TABLE 4
TRANSFORMER AND RECTIFIER PARAMETERS

| Quantity | Symbol | Value |
|---|---|---|
| Mains voltage (RMS) | $V_p$ | 220 V |
| Mains peak voltage | $V_{p,\mathrm{pk}}$ | 311.1 V |
| Secondary voltage (RMS, measured) | $V_s$ | 10.53 V |
| Secondary peak voltage | $V_{s,\mathrm{pk}}$ | 14.89 V |
| Diode forward drop | $V_D$ | 0.7 V |
| Rectifier peak output | $V_{o,\mathrm{pk}}$ | 13.49 V |
| Rectifier average output | $V_{o,\mathrm{avg}}$ | 8.59 V |
| Rectifier RMS output | $V_{o,\mathrm{rms}}$ | 9.54 V |
| Mains frequency | $f$ | 50 Hz |
| Rectifier ripple frequency | $f_r$ | 100 Hz |
| Transformer turns ratio | $N_p : N_s$ | 20.9 : 1 |
| Peak inverse voltage (approx.) | PIV | 14.89 V |
| 1N4007 rating | – | 1000 V, 1 A |
| 1N5408 rating | – | 3 A |

The corresponding first-order RC transfer function is

$$H(s) = \frac{V_C(s)}{V_{\mathrm{in}}(s)} = \frac{1}{1 + sR_LC}, \qquad \tau = R_LC. \quad (6.11)$$

Here, $V_{\max}$ is approximately the rectifier peak output. Since the load current is not yet measured, the ripple is estimated symbolically. For small ripple and nearly constant load current,

$$V_{r(pp)} \approx \frac{I_L}{f_rC} = \frac{I_L}{100(2.2 \times 10^{-3})} \quad (6.12)$$

$$\approx 4.545 I_L \text{ V}, \quad (6.13)$$

where $I_L$ is in amperes. Thus,

$$V_{DC} \approx V_{\max} - \frac{V_{r(pp)}}{2} \quad (6.14)$$

$$\approx 13.49 - 2.273 I_L \text{ V}. \quad (6.15)$$

These are theoretical estimates; the exact ripple and time constant depend on the load.

### *6.5 Regulator*

#### *6.5.1 IC 7805 Regulator:*

#### *6.5.2 LM2596 Buck Converter:*

### *6.6 USB Port*

The D+ and D- pins are used by legacy USB charging systems to identify the source's charging capability. The device negotiates or detects the available current; the charger does not force that current into the device. The following values are reference predictions and depend on device compatibility.

The BC1.2 values are standard charging-port limits. The Apple voltage pairs are legacy proprietary signatures and are not universally supported. USB Type-C and USB Power Delivery use different identification and negotiation methods; D+/D- voltage pairs alone must not be treated as a universal current-control method.

*References:* USB-IF, *Battery Charging Specification, Revision 1.2*; Microchip, *Battery Charging with USB*; Texas Instruments, *USB Battery Charging Specification 1.2 (BC1.2)*; and the legacy Apple charging-voltage summary in https://verasol.org/wp-content/uploads/import/Issue-24_USB-smartphone-charging-final.pdf.

TABLE 4
USB D+/D- Charging Signatures and Reference Current Limits

| **Charging mode** | **D+** | **D-** | **Reference current li** |
|---|---|---|---|
| USB 2.0 SDP | Standard data-line pull-downs | Standard data-line pull-downs | 500 mA after configura |
| USB 3.x SDP | Standard data-line pull-downs | Standard data-line pull-downs | 900 mA after configura |
| BC1.2 CDP | Detection handshake | Detection handshake | Up to 1.5 A |
| BC1.2 DCP | Shorted to D- | Shorted to D+ | Up to 1.5 A |
| Legacy Apple signature | 2.0 V | 2.0 V | Approx. 500 mA |
| Legacy Apple signature | 2.0 V | 2.7 V | Approx. 1.0 A |
| Legacy Apple signature | 2.7 V | 2.0 V | Approx. 2.1 A |
| Legacy Apple signature | 2.7 V | 2.7 V | Approx. 2.4 A |